

Marking instructions BAN404

Instructions

Your solution should be submitted as two files.

1. A pdf-file where both your R-code and the answers to the tasks are given.
2. Either a well documented R-script (.R) or an R markdown file (.Rmd) which can reproduce the results in the pdf-file.

The datasets used here are well analyzed before. Make sure to refer to other's work if you use it to solve the tasks.

A general recommendation: Be pragmatic if you run into difficulties. You need to find models that can produce predictions. Don't let the perfect prediction stand in the way for a good one!

The number in brackets after each subtask shows the maximum score for that subtask.

R coding can be done in different ways than done here, e.g. using the tidyverse approach. Given a correct and clearly written result there is no preference for any particular approach.

There were 4 versions of the exam where the seeds for the random numbers were different in subtasks 1a), 1e) and 2a) . Here only one version is shown.

Throughout the exam you will work with training and test data and you will have to make appropriate choices on when to use one or the other. In cases when you think there are more than one valid choice on working with the training, test or the full dataset, explain your choice!

Task 1

In this task you will look at the OJ data in the ISLR package. The dataset contains 1070 purchases of two brands of juice. In the task you should analyze and predict the variable **Purchase**, a variable showing which of the two brands, Citrus Hill (CH) or Minute Maid Orange (MM), was purchased. See the help-function to the dataset for more information of the dataset. Except in Task e), you should assume that the variable **LoyalCH**, a variable indicating loyalty to the Citrus Hill brand, is unavailable at the time of prediction.

- a) Remove the variable **LoyalCH**. If necessary, recode categorical variables as factors and remove variables that cannot be used in the analysis. Base your reasoning to do this on the help function and by looking at the data. Split the data in a 50/50 split to create a training and a test dataset. Use the seed 8655, when you split the data, `set.seed(8655)`

Suggested solution:

```
library(ISLR) # Read library and data
# Remove LoyalCH and StoreID, which the same as STORE
xdata=OJ[,!(names(OJ) %in% c("LoyalCH","StoreID"))]
# Compute how many possible values for the variables
# If few values, consider to recode as factor in cases where the
# variable does not have a natural quantitative interpretation.
card=function(x) length(table(x))
len=sapply(xdata,card)
len

##      Purchase WeekofPurchase      PriceCH      PriceMM      DiscCH
##           2           52           10           8           12
##      DiscMM      SpecialCH      SpecialMM      SalePriceMM      SalePriceCH
##           12           2           2           18           13
##      PriceDiff      Store7      PctDiscMM      PctDiscCH      ListPriceDiff
##           36           2           18           13           18
##      STORE
##           5

#sapply(xdata,class)
for(i in 1:length(len)) if(len[i]<=5) xdata[,i]=as.factor(xdata[,i])
#sapply(xdata,class)
n=nrow(xdata)
ntrain=floor(n/2)
set.seed(v1)
ind=sample(1:n,ntrain)
train=xdata[ind,]
test=xdata[-ind,]
```

- b) Use descriptive statistics to detect promising predictors for **Purchase**. Write in words which R functions you used, present the most interesting results as tables and graphs and comment on them.

Suggested solution:

```
par(mfrow=c(1,2))
plot(Purchase~STORE,data=train)
boxplot(PriceDiff~Purchase,data=train)
```



```
by(train$PctDiscCH,train$Purchase,mean)
```

```
## train$Purchase: CH
## [1] 0.03790963
## -----
## train$Purchase: MM
## [1] 0.01590576
```

```
by(train$PctDiscMM,train$Purchase,mean)
```

```
## train$Purchase: CH
## [1] 0.04600432
## -----
## train$Purchase: MM
## [1] 0.08483992
```

First plotted **Purchase** against all other variables with `plot(Purchase~.,data=train)`. Furthermore means of numerical variables for the two values of **Purchase** was computed. A variable that stand out as promising are **STORE** with a large variation on which brand is bought. Another is **PriceDiff** which, on average, is higher in cases when the, usually, cheaper CH, was bought. A third promising variable is **PctDiscCH**, the discount on CH; and similarly for **PctDiscMM**. The variable **SpecialMM** also is differently distributed between the between the two brands.

- c) Fit a logistic regression of **Purchase** on all the variables you found promising in 1b, evaluate the predictions using test accuracy and a confusion matrix. If you, in the process, detect that you would modify your model, you should do so and explain why. Use the threshold 0.5 when you go from estimated probability to a prediction.

Suggested solution:

```
logit1=glm(Purchase~PriceDiff+STORE
           ,data=train,family=binomial())
summary(logit1)

##
## Call:
## glm(formula = Purchase ~ PriceDiff + STORE, family = binomial(),
##      data = train)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9947  -0.8912  -0.5919   1.0260   2.2060
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.7678     0.1826  -4.204 2.62e-05 ***
## PriceDiff    -2.4588     0.3975  -6.186 6.16e-10 ***
## STORE1        0.6979     0.3059   2.282  0.0225 *
## STORE2        1.1550     0.2658   4.346 1.39e-05 ***
## STORE3        1.8724     0.2945   6.357 2.05e-10 ***
## STORE4       -0.3935     0.3551  -1.108  0.2678
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 714.98  on 534  degrees of freedom
## Residual deviance: 606.08  on 529  degrees of freedom
## AIC: 618.08
##
## Number of Fisher Scoring iterations: 4

pred.logit1=predict(logit1,type="response",newdata=test)
table(test$Purchase,pred.logit1>0.5)

##
##      FALSE TRUE
## CH    261    65
## MM     86   123

tab1=round(prop.table(table(test$Purchase,pred.logit1>0.5),margin=1),3)
tab1

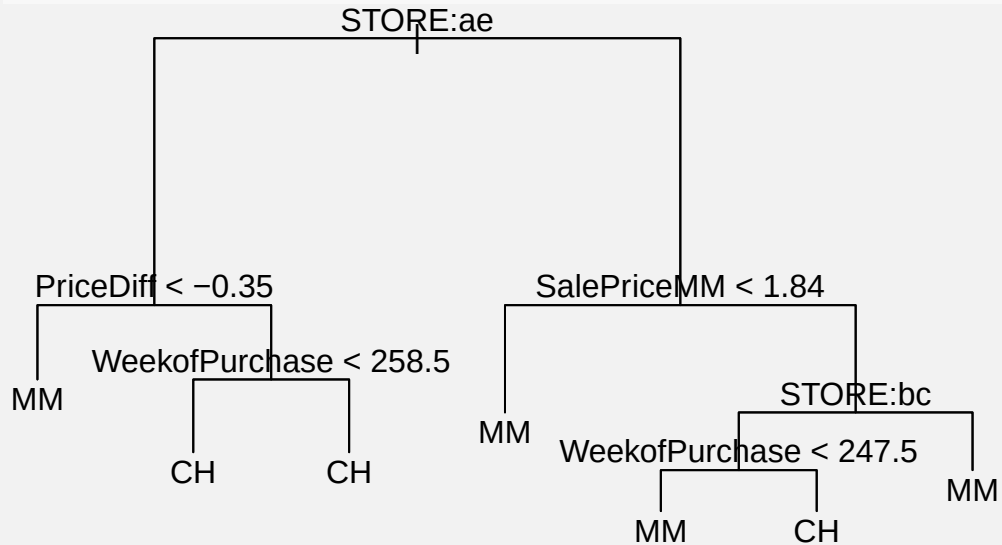
##
##      FALSE TRUE
## CH 0.801 0.199
## MM 0.411 0.589
```

Since the discount variables and SpecialMM were insignificant in the logistic regressions they were removed. The model predicts 80% of the CH purchases correctly and 59% of the MM purchases.

- d) Fit a classification tree, using all variables (except the one(s) you removed in 1a), and plot it. Give an example of how to interpret a branch of the tree (How to predict a new observation with certain values on the explanatory variables). Also, use the same evaluation measures as in 1c and evaluate the predictions. Use the threshold 0.5 when you go from estimated probability to a prediction.

Suggested solution:

```
library(tree)
tree1=tree(Purchase~.,data=train)
plot(tree1)
text(tree1)
```



```
prob.tree1=predict(tree1,newdata=test)
pred.tree1=prob.tree1[,2]>0.5
table(test$Purchase,pred.tree1)
```

```
##      pred.tree1
##      FALSE TRUE
##  CH    235    91
##  MM     65   144
```

```
round(prop.table(table(test$Purchase,pred.tree1),margin=1),3)
```

```
##      pred.tree1
##      FALSE TRUE
##  CH 0.721 0.279
##  MM 0.311 0.689
```

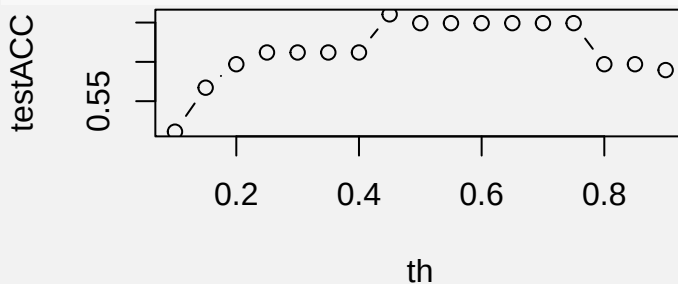
```
ctable=table(test$Purchase,pred.tree1)
testACC=sum(diag(ctable))/sum(ctable)
```

Example of a prediction: A purchase in store a or e (0 or 4) with a price difference larger than 0.085 is likely to be of the, mostly, cheaper brand Citrus Hill.

- e) In the 1c and 1d you have used the threshold 0.5. Use cross-validation to find the threshold that maximizes the accuracy, i.e. the fraction of correctly predicted observations. Evaluate predictions from the tree using the best threshold. Are the predictions improved? Use the seed 4598, when you split the data, `set.seed(4598)`.

Suggested solution:

```
set.seed(v2)
ntrain2=floor(ntrain/2)
ind2=sample(1:ntrain,size=ntrain2)
train2=train[ind2,]
test2=train[-ind2,]
tree2=tree(Purchase~.,data=train2)
testACCvec=data.frame(th=0,testACC=0)
thval=seq(0.1,0.9,0.05)
k=1
for(th in thval)
{
  pred.tree2=predict(tree2,newdata=test2)[,2]>th
  ctable=table(test2$Purchase,pred.tree2)
  testACC=sum(diag(ctable))/sum(ctable)
  testACCvec[k,]=cbind(th,testACC)
  k=k+1
}
plot(testACCvec,type="b")
```



```
prob.tree1=predict(tree1,newdata=test)
thopt=thval[which.max(testACCvec[,2])]
pred.tree1=prob.tree1[,2]>thopt
table(test$Purchase,pred.tree1)
```

```
##      pred.tree1
##      FALSE TRUE
## CH    235    91
## MM     65   144
```

```
round(prop.table(table(test$Purchase,pred.tree1),margin=1),3)
```

```
##      pred.tree1
##      FALSE TRUE
## CH 0.721 0.279
## MM 0.311 0.689
```

```
ctable=table(test$Purchase,pred.tree1)
testACC=sum(diag(ctable))/sum(ctable)
testACC
```

```
## [1] 0.7084112
```

For logistic regression a very similar code can be used where the logistic regression is fitted instead of classification trees and the prediction function is augmented with the argument `type="response"` and result in a n -vector rather than a $n \times 2$ -matrix.

- f) In a) you removed the variable `LoyaltyCH`. Imagine now that this variable is actually available at the time of a prediction. By augmenting your analysis with this variable both for the logistic regression

and the classification tree, conclude if this variable contribute to predicting Purchase.

Suggested solution: Rerun the analysis in a)-e) but do not remove LoyaltyCH in a). Compare confusion matrices and test accuracies.

Task 2

In this task you will work with the dataset `Computers` in the `Ecdat` package. You should predict the variable `price` using properties of the computers and other explanatory variables. In the process of this we would also like to learn something about the properties of low- and high-priced computers.

- a) Just as in 1a; if necessary, recode categorical variables as factors and remove variables that cannot be used in the analysis. Base your reasoning to do this on the help function and by looking at the data. Split the data in a 50/50 split to create a training and a test dataset. Use the seed 4598, when you split the data, `set.seed(4598)` (3)

Suggested solution:

```
library(Ecdat)

## Loading required package: Ecdat
##
## Attaching package: 'Ecdat'
## The following object is masked from 'package:base':
##
##     sign
##
## Attaching package: 'Ecdat'
## The following object is masked from 'package:datasets':
##
##     Orange

data(Computers)
len=apply(Computers,card)
len
##   price   speed    hd    ram  screen    cd  multi premium   ads  trend
##    808     6    59     6     3      2    2      2      34    35

#apply(Computers,class)
for(i in 1:length(len)) if(len[i]<=5) Computers[,i]=as.factor(Computers[,i])
#apply(Computers,class)
n=nrow(Computers)
ntrain=floor(n/2)
set.seed(4598)
ind=sample(1:n,ntrain)
train=Computers[ind,]
test=Computers[-ind,]
```

- b) Use descriptive statistics to find promising predictors. Comment on your observations. (7)

Suggested solution:

```
plot(price~.,data=train)
```

- c) Run a linear regression on all explanatory variables. Interpret some of the coefficients. Evaluate the

prediction performance on an appropriate measure. (14)

Suggested solution:

```
reg1=lm(price~.,data=train)
summary(reg1)

##
## Call:
## lm(formula = price ~ ., data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1092.33  -175.81    -8.33   144.98  1952.13
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2020.70893   30.33070   66.623 < 2e-16 ***
## speed         9.34193    0.25816   36.187 < 2e-16 ***
## hd            0.80181    0.03867   20.736 < 2e-16 ***
## ram          49.09392    1.50968   32.519 < 2e-16 ***
## screen15     50.62973   11.76869    4.302 1.74e-05 ***
## screen17    399.06831   16.82983   23.712 < 2e-16 ***
## cdyes        59.16985   13.52546    4.375 1.26e-05 ***
## multiyes    109.11924   16.33084    6.682 2.78e-11 ***
## premiumyes -500.65735   16.92124  -29.588 < 2e-16 ***
## ads          0.65687    0.07381    8.900 < 2e-16 ***
## trend      -51.56800    0.89310  -57.741 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 276.7 on 3118 degrees of freedom
## Multiple R-squared:  0.781, Adjusted R-squared:  0.7803
## F-statistic: 1112 on 10 and 3118 DF, p-value: < 2.2e-16

pred1=predict(reg1,newdata=test)
MSE1=mean((test$price-pred1)^2)
MSE1
## [1] 72200
```

- d) Investigate if the the predictions of `price` would be better if we use a linear regression of `log(price)` on the other variables. (7)

Suggested solution:

```
Computers$logp=log(Computers$price)
train=Computers[ind,]
test=Computers[-ind,]
reg2=lm(logp~.-price,data=train)
pred2=exp(predict(reg2,newdata=test))
MSE2=mean((test$price-pred2)^2)
MSE2
## [1] 72610.7
```

- e) In 2d you fitted a model where `price` is a particular nonlinear function of the other variables. You

should now investigate other non-linear models. First, fit a GAM-model, plot the result and evaluate the predictions. Is there a reason to not allow for some variables to have a non-linear relationship with price? (4)

Suggested solution:

```
library(gam)
```

```
## Loading required package: splines
```

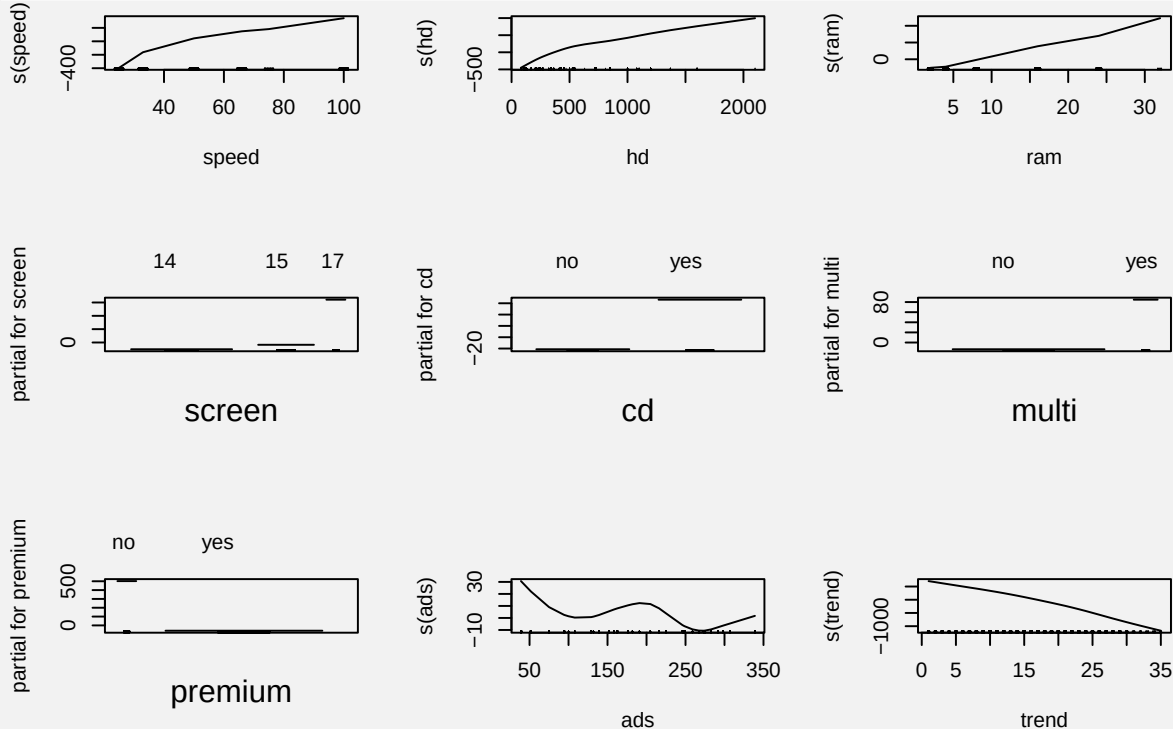
```
## Loading required package: foreach
```

```
## Loaded gam 1.22-1
```

```
gam1=gam(price~s(speed)+s(hd)+s(ram)+screen+cd+multi+premium+s(ads)+s(trend),data=train)
```

```
par(mfrow=c(3,3))
```

```
plot(gam1)
```



```
pred3=predict(gam1,newdata=test)
```

```
MSE3=mean((test$price-pred3)^2)
```

```
MSE3
```

```
## [1] 60947.91
```

```
Improved the predictions quite a lot.
```

- f) Give a brief explanation of how backfitting works and in what situations it is possible to fit a generalized additive model with ordinary least square regression.

Suggested solution: In backfitting a model of the form (example with two predictors)

$$y = f_1(x_1) + f_2(x_2) + \varepsilon$$

is estimated by first setting $f_2(x_2) = 0$ and estimate f_1 with a method for one-predictor models, e.g., the Nadaraya-Watson estimator. The residuals $e_1 = y - \hat{f}_1(x_1)$ are then computed and will substitute the response variable in the next iteration where e is fitted to $f_2(x_2)$. Now, the residuals $e_2 = e_1 - \hat{f}_2(x_2)$ are computed and again fitted to $f_1(x_1)$. This procedure is continued until the estimators of f_1 and f_2 converges.

A GAM can be fitted with OLS if the functions $f_k(x_k)$ can be written with so called basis functions. Examples of that are splines and piecewise polynomials.

- g) Use bagged trees to predict **price**. Evaluate the predictions. Compute variable importance measures and comment on the results. Explain the logic behind the measure for variable importance that you are using. (Hint: If the computation takes a long time you can reduce the number of trees fitted using the argument **ntree**.)

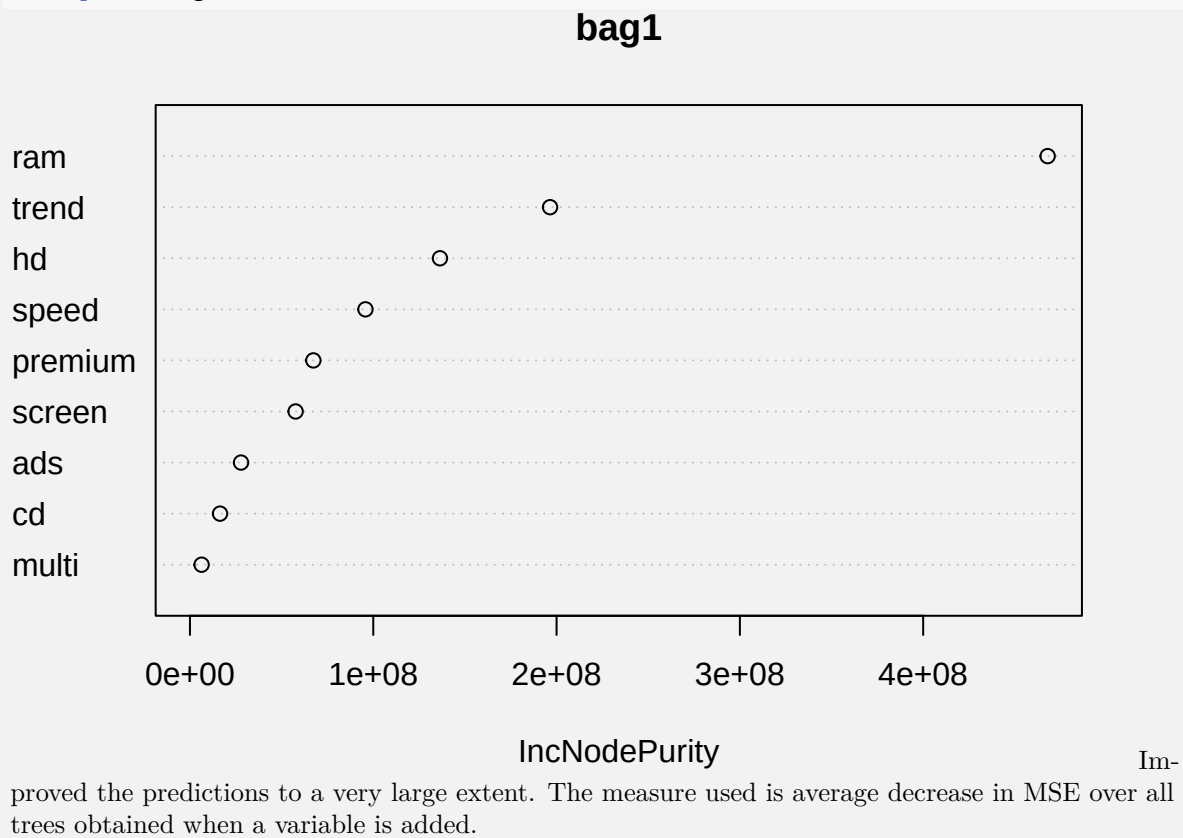
Suggested solution:

```
library(randomForest)

## randomForest 4.7-1.1
## Type rfNews() to see new features/changes/bug fixes.

bag1=randomForest(price~.-logp,data=train,mtry=9,ntree=100)
pred4=predict(bag1,newdata=test)
MSE4=mean((test$price-pred4)^2)
MSE4
## [1] 28272.51

varImpPlot(bag1)
```



h) Give a brief explanation of bagging.

Suggested solution: In bagging, bootstrap samples (a draw, with replacement, of the same size as the original sample) are used to fit a model, e.g., a regression tree. A prediction is done by the model for each bootstrap sample and the final prediction is computed by averaging all of them.